

JOINTS

THE SITE AT WHICH TWO BONES LINK IS CALLED A JOINT OR AN ARTICULATION. JOINTS CAN BE CLASSIFIED ACCORDING TO THEIR STRUCTURE AND BY THE TYPES OF MOVEMENT THEY ALLOW. THE BODY HAS MORE THAN 300 DIFFERENT JOINTS.

SYNOVIAL JOINTS

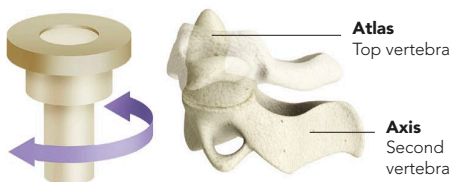
The body's most numerous, versatile, and freely moving joints are known as synovial joints. They can work well for many decades if kept in good use, but not overused. Synovial joints are enclosed by a protective outer covering – the joint capsule. The capsule's inner lining, called the synovial membrane, produces slippery, viscous synovial fluid that keeps the joint well lubricated so that the joint surfaces slide against each other with minimal friction and wear. There are around 230 synovial joints in the body.

TYPES OF SYNOVIAL JOINT

A synovial joint's range of movement is determined by the shape of its articular cartilage surfaces (see p.46) and how they fit together.

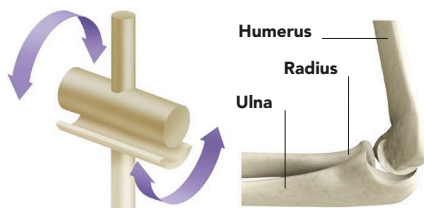
Pivot joint

A peg-like projection from one bone turns in a ring-shaped socket of another bone; or, conversely, the ring turns around the peg. The pivot joint between the top two neck (cervical) vertebrae enables the skull to rotate on the spinal axis, and to move from side to side.



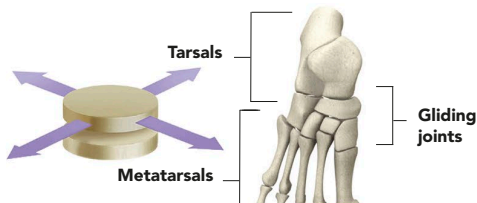
Hinge joint

The convex surface of one bone fits into the concave surface of another bone to allow to-and-fro movement, mainly in one plane. The elbow is a modified hinge joint that permits limited rotation of the arm bones.



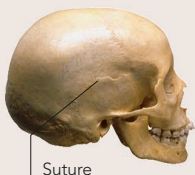
Gliding joint

The bone surfaces that meet in a gliding joint are almost flat and slide over one another. Movement is limited by ligaments. Some joints between the tarsals of the ankle and between the carpals in the wrist move in this way.



SEMIMOVABLE AND FIXED JOINTS

Not all joints have a wide range of movement. Some allow for growth or for greater stability. The bones in these joints are usually linked by cartilage or tough fibres made of substances such as the protein collagen. In the fixed joints of the skull, once growth is complete, the separate bone plates are securely connected by interlocking fibrous tissue, forming suture joints.



FIXED JOINT

The adult skull's suture joints show up as wiggling lines. In infancy, these joints are loosely attached to allow for expansion of the rapidly growing brain.

SEMIMOVABLE JOINT

In partly flexible joints, bones are linked by fibrous tissue or cartilage, as in the pubic symphysis.

Pubic symphysis

